

Q1. There are 4 Pillars of OOPs as follows;

1. Inheritance:

- i. Single Inheritance
- ii. Multiple Inheritance
- iii. Multilevel Inheritance
- iv. Hierarchical Inheritance
- v. Hybrid Inheritance

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2. Encapsulation:

- i. Public
- ii. Private
- iii. Protected

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3. Polymorphism:

- i. Overloading
- ii. Overriding

4. Data Abstraction

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Q2. Overloading:

is defined as accumulation of data and data type, as the name explains it accumulates multiple data and its type whether with same operator or multiple operator in a single class.

* Child class possesses, same operator behaves differently as per the Data type.

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Overriding:

is overwriting or modifying the parent class to a new child class which possesses updated advanced or new logic than the parent class.

* It cannot be achieved in a single class as overloading does as it requires two or more/multiple class to assign the modification or change in the child class.

* In run time Python decides which method to execute.

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Q3. Encapsulation:

Encapsulation means wrapping of data and method together in a single unit(class) and restricting direct access to the data.

Encapsulation = Data Hiding + Data protection

Without Encapsulation data is unsafe, no control, have bugs, etc.

With Encapsulation data is Protected, Controlled, Secured, etc.

Example : ATM Machine: => Cannot access data/account directly. => Need a method to access. => Balance is hidden.

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Q4. Types of Inheritance:

- i. Single Inheritance
- ii. Multiple Inheritance
- iii. Multilevel Inheritance
- iv. Hierarchal Inheritance
- v. Hybrid Inheritance

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Q5. Types of case in programming language:

Types of Case	Example	Description
camelCase	studentData	First word starts with lowercase and following each word starts with a capital letter.
PascalCase	StudentData	Each word starts with a capital letter along with the first word
snake_case	student_data	words are in lowercase seperated by underscore sign (_)

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Q6. Class: A class is defined as a Blueprint or Template that is used to create objects in OOPs.

Key Points:

- * Is defined using class keyword
- * It contains variables and method
- * Variables are written in PascalCase

Q7. Object: An object is an instance of any class.

syntax: s1 = Student()

Key Points:

- > Memory is allocated as soon as the object is created.
- > It can access class method and variables.

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Q8. Types of Variables in OOPs:

- 1. Instance Variable: defined using (self)
- 2. Class Variable: shared among all objects

example: self.name = "ajay"

example: college = "MIT,Pune"

Q9. Types of Methods used in OOPs:

1. Instance Method ✓
2. Class Method ✓
3. Static Method ✓

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Q11. Rules of Overriding:

1. Parent and child method names must be same. ✓
2. Signature should be same i.e same Method name and Parameters. ✓
3. Overriding is possible only if there is an inheritance between parent and child class.
4. As the name suggests child method overwrites/overrides the parent method and only child method is executed. ✓

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Q12. Super in OOPs:

- * Super is used to call the Parent class method in the child class.
- * Thus it has features of Parent class as well as added/updated functions of the child class in it. ✓

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Q14. Steps to write a file using Python:

1. Open file: Open() ✓
2. Write data into the file.: write() ✓
3. Close the file.: close() ✓

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Q18. Difference between 'w' and 'x' in writing files:

- * 'w' stands for "write" that creates a file or overwrites the existing file whereas;
- * 'x' stands for "create" in which a file is created i.e. new file, or error if it exists. ✓

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Q19. Lambda Function :

A lambda function is an anonymous(Unnamed) function.

Methods of lambda:

1. map()
2. filter()
3. reduce()
4. apply()



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Q21.

L = "Comprehension"
[ch for ch in L if ch in "aeiouAEIOU"]



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